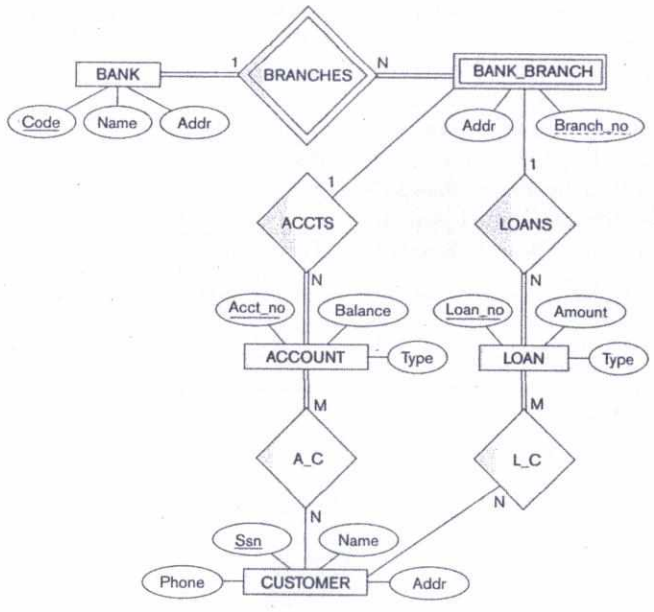
Roll Number: _____ Name_____ Group_____

THAPAR INSTITUTE OF ENGINEERING &TECHNOLOGY, PATIALA
Department of Computer Science & Engineering
Database Management System (UCS310) – Mid Semester Examination
Date: 07/03/2025 & Time: 11:00AM **MM: 30 & MT: 120 Min**

***Faculty Names**: Dr Geeta Kasana, Dr. Garima Singh, Dr Shashank Singh, Dr. Amrita Dahiya, Dr. Swati Sharma, Dr. Tanya Garg, Dr. Ananya Pandey, Ms. Shivani Goswami, Dr. Manisha Kaushal.*

***Attempt/Answer all sub-parts like (a), (b) for each question at one place. Do mention Page No. of your attempt on the front page of the answer sheet.** Assume missing data (if any). Show all intermediate computations properly.*

Q. No.		Question	Marks	BL	CLO
	(a)	An online shopping platform initially managed its customer orders using a file-based system, where separate files stored customer details, product inventory, orders, and payments. As the business grew, they faced issues in updating stock levels, tracking order status, and managing customer details efficiently. To overcome these problems, they switched to a Database Management System (DBMS). Compare File-Based System and DBMS for the e-commerce business based on: i) Data Consistency ii) Data Redundancy iii) Concurrency Handling (Multiple Users Accessing Data Simultaneously) iv) Data Integrity in Order Processing	4	L4	1
	(b)	Consider a university that wants to develop a database management system to handle student records, courses, and faculty details. Different users, such as students, professors, and administrators, require access to specific data while maintaining security and abstraction. Explain how the three-level architecture of DBMS (Internal, Conceptual, and External levels) can be applied to this scenario.	3	L4	1
2	(a)	 i) List the Primary keys of All the relations. ii) Describe the banking scenario depicted by this E-R diagram in detail. iii) Find the minimum number of tables required to represent the above ER- Diagram	4	L4	2

		iv) Suppose that every customer must have at least one account but is restricted to at most two loans at a time, and that a bank branch cannot have more than 1,000 loans. How does this show up on the (min, max) representation?			
	(b)	Explain referential integrity constraint that must be maintained during insertion, deletion, and updating of records in tables. How do cascading actions affect data integrity in relational databases?	3	L2	2
Q3	(a)	Consider a relation R (A, B, C, D, E, F) with the following functional dependencies (FDs): A→BC CD→E B→D E→A F→B The relation R is decomposed into three sub-relations: R1(A, B, C), R2(A, D, E), R3(B, F) Analyze whether the given decomposition is: i) Lossy or Lossless ii) Dependency-Preserving or Not	3	L5	3
	(b)	Consider Schema R1 and R2, R1 holds FD1, and R2 holds FD2: **R1(Order_ID, Order_Date, Customer_ID, Customer_Name, Customer_Address, Product_ID, Product_Description, Product_Finish, Unit_Price, Order_Quantity)** **FD1:** Order_ID → Order_Date, Customer_ID, Customer_Name, Customer_Address Customer_ID → Customer_Name, Customer_Address Product_ID → Product_Description, Product_Finish, Unit_Price Order_ID, Product_ID → Order_Quantity **R2(Order_ID, Order_Date, Customer_ID)** **FD2:** Order_ID, Order_Date → Customer_ID Customer_ID → Order_Date For the given schema R1 and R2: i) Identify the candidate keys for the relation R1 and R2. ii) Identify Prime and Non-Prime Attributes of R1 and R2. iii) Determine the highest normal form of Relation R1 and R2. iv) Normalize R1 and R2 both upto BCNF and show all decomposed tables at each level with key attributes.	5	L5	3
Q4	(a)	Consider two relations EMP(Employee_ID, Employee_Name, Department, Salary) and Dept(Department, Location). Write SQL queries for following: i) Find Employees Whose Departments Have No Assigned Location ii) Find Employees Who Work in the Same Department iii) Find departments Name where the average salary is greater than 80,00 iv) Find Departments Name Where Employee Count is More Than 1 v) Modify table to ensure salary must be greater than 30000. vi) Find out Employee Names working on same Location.	6	L5	5
	(b)	Identify which SQL clause belongs to **DML, DDL, and DQL**. Create a table with three columns: **DDL, DML, and DQL**, and categorize the given SQL commands accordingly. DROP, RENAME, UPDATE, DELETE, SELECT, TRUNCATE , ALTER, INSERT	2	L2	5

Employee_ID	Employee_Name	Department_ID	Salary
101	Alice	1	70000
102	Bob	2	90000
103	Charlie	3	85000
104	David	2	95000
105	Emma	4	65000
106	Frank	1	72000

Department_ID	Department_Name	Location
1	HR	New York
2	IT	Chicago
3	Finance	Chicago
5	Marketing	Los Angeles